

Foundations of Trustworthy AI

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Class 6: Omniprediction

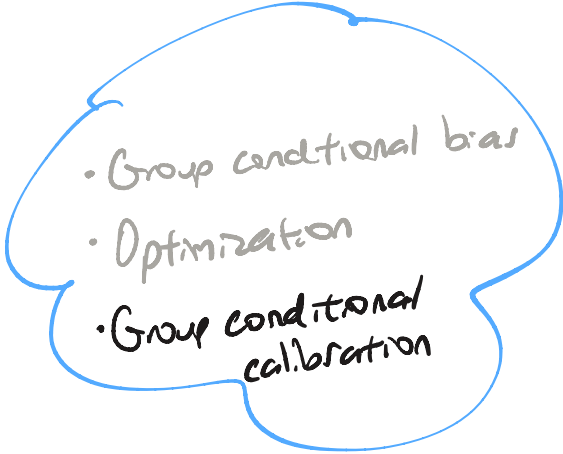
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Roadmap

Calibration topics:

- Prediction, bias, and calibration
- Measuring calibration error
- Decision theoretic implications
- Group conditional calibration
- Omniprediction
- Conformal prediction

... then onto "Robustness"

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- Group conditional bias
 - Optimization
 - Group conditional calibration

Recap

Definition: For a distribution D , predictor f , and group g ,
the **group conditional bias** is

$$\beta_D(f, g) := \mathbb{E}_D[f(x) - y \mid g(x) = 1]$$

Definition: For a distribution D , predictor f , and group class G ,
 f is **α -group class unbiased** if

$$\forall g \in G \quad \beta_D(f, g)^2 \mu_D(g) \leq \alpha^2 \quad \text{where } \mu_D(g) = \mathbb{P}_D[g(x) = 1]$$

Recap

Algorithm for reducing bias

Input: f, G, α

1. Set $f_0 = f$ and $t = 0$
2. While there is $g_t \in G$ st.

$$\beta(f_t, g_t)^2 \mu(g_t) > \alpha^2$$

set

$$f_{t+1}(x) = f_t(x) - \beta(f_t, g_t) g_t(x)$$

increment t

Output: f_t

Definition: For a distribution D , predictor f , and group class G , f is α -group class unbiased if

$$\forall g \in G \quad \beta_D(f, g)^2 \mu_D(g) \leq \alpha^2 \quad \text{where } \mu_D(g) = \mathbb{P}_D[g(x)=1]$$

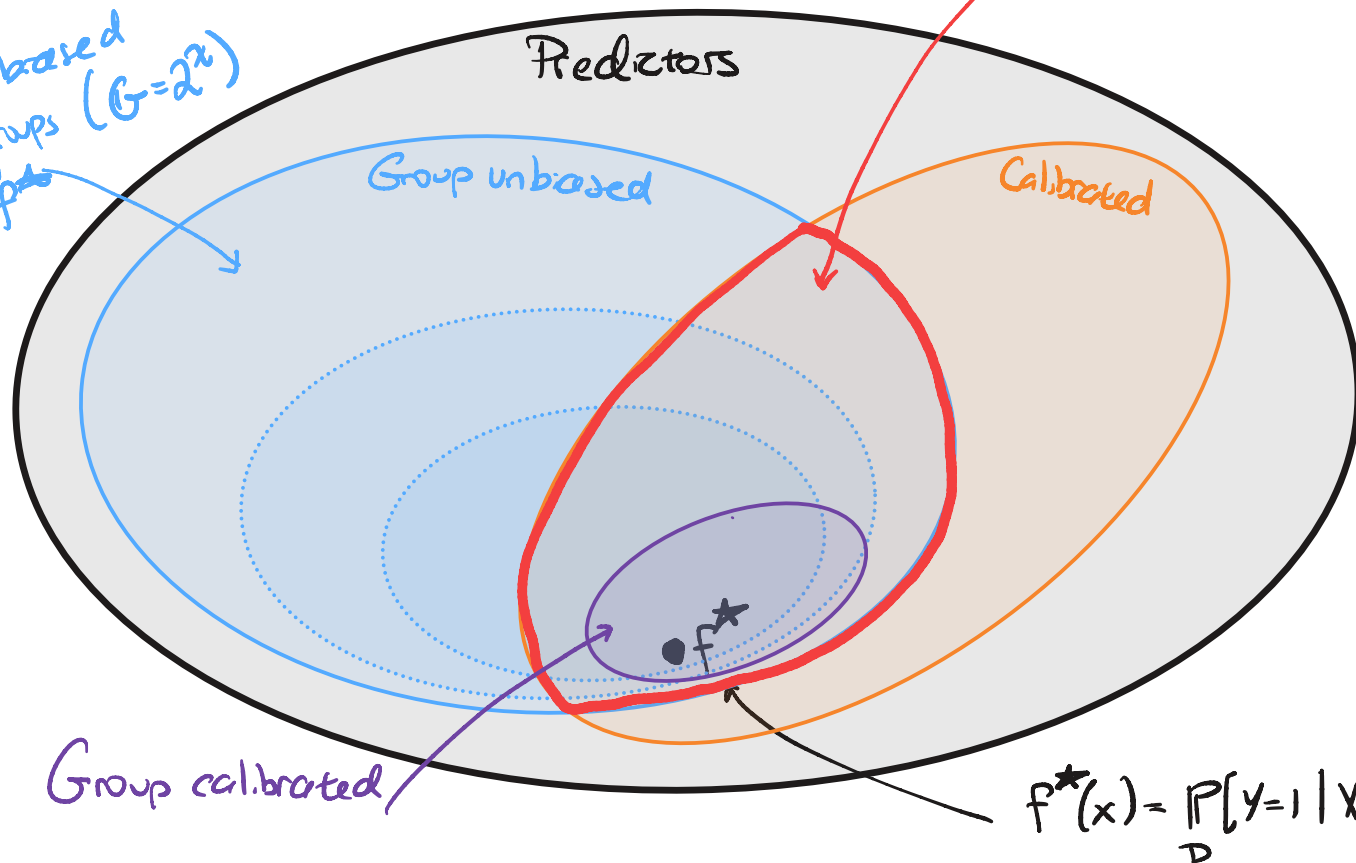
Lemma: If $f'(x) = f(x) - \beta(f, g) g(x)$
then $MSE(f') = MSE(f) - \beta(f, g)^2 \mu(g)$

Technique: Use $MSE(f_t)$ as a potential

Theorem: After $t \leq 1/\alpha^2$ steps
the algorithm outputs f_t
that is α -unbiased for G

Group Conditional Calibration

Group unbiased
for all groups ($G=2^x$)
means $f = f^*$



Group Class Unbiased

For a group g and prediction $p \in (0, 1]$ define

$$g_{f,p}(x) = g(x) \mathbb{1}_{\{f(x)=p\}} \quad "g_{f,p} \text{ is 1 if } x \text{ in the group and } f(x) = p"$$

$$\beta(f, g_{f,p}) = \mathbb{E}[f(x) - y \mid g(x)=1, f(x)=p]$$

$$\mu(g_{f,p}) = \mathbb{P}[g(x)=1, f(x)=p]$$

Definition: For distribution $\mathcal{D}$, predictor f , group class G , f is α -group calibrated if

$$\forall g \in G \sum_{p \in \text{range}(f)} \beta(f, g_{f,p})^2 \cdot \mu(g_{f,p}) \leq \alpha^2$$

Aside

$$\sum_{p \in \text{range}(f)} \mathbb{P}[f(x)=p] \cdot |p - \mathbb{E}[y \mid f(x)=p]|$$

Recap

calibration error

Algorithm for reducing ~~bias~~

Input: f, G, α

1. Set ~~$P_0 = f$~~ and $t=0$ $m = 1/\alpha^2$ $f_0 = \text{Round}(f; m)$ Note: rounding can increase MSE by $\leq 1/2m$

2. While there is $g_t \in G$ st.

$$\beta(f_t, g_t)^2 \mu(g_t) > \alpha^2 \quad \beta(f_t, g_t)_{f_t, p_t}^2 \mu(g_t)_{f_t, p_t} > \alpha^2 / m = \alpha^4$$

set

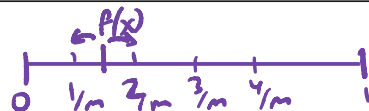
$$\cancel{f_{t+1}(x) = f_t(x) - \beta(f_t, g_t) g_t(x)}$$

increment t

Output: f_t

Definition: For distribution D , predictor f , group class G , f is α -group calibrated if

$$\forall g \in G \sum_{p \in \text{range}(f)} \beta(f, g)_{f, p}^2 \mu(g)_{f, p} \leq \alpha^2$$



$$q_t = \text{Round}(\mathbb{E}_D[Y | g_t(x)=1, f_t(x)=p_t]; m)$$

$$f_{t+1}(x) = \begin{cases} q_t & \text{if } \underline{g_t(x)=1, f_t(x)=p_t} \\ f_t(x) & \text{o.w. } (g_t)_{f_t, p_t}(x)=1 \end{cases}$$

Omnipredictors

Loss function: $l(p, y)$

All of ML

Optimization / Model Fitting:

$$\operatorname{argmin}_{h \in H} \mathbb{E}_{\mathcal{D}} [l(h(x), y)]$$

Examples of loss functions

$$l(p, y) = (p - y)^2$$

$$l(p, y) = |p - y|$$

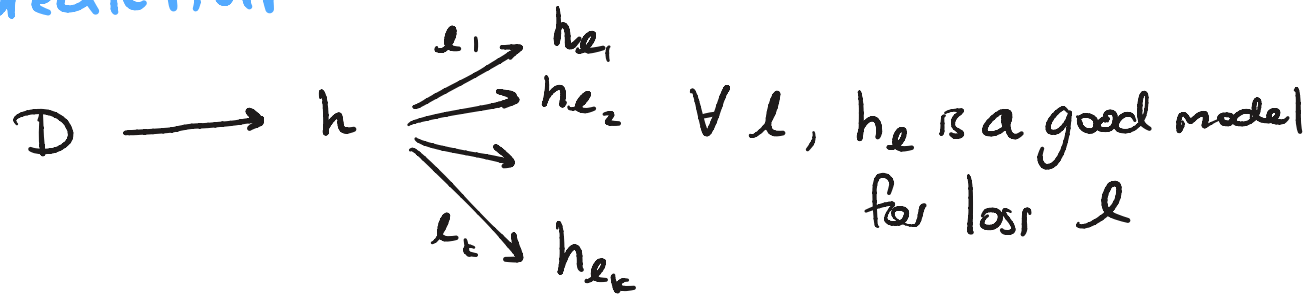
Even if $f^*(x) = \mathbb{P}[Y=1 | X=x]$ is known

Even if $H = \{\text{all fns}\}$

Then different losses lead to different models

$$\begin{aligned} & \begin{array}{l} (p-y)^2 \rightarrow \\ |p-y| \rightarrow \end{array} h = f^* \\ & h = \begin{cases} 1 & \text{if } f^* \geq 1/2 \\ 0 & \text{if } f^* < 1/2 \end{cases} \end{aligned}$$

Omni:prediction



Omniprediction

A ~~utility~~^{loss} function ~~$U(a, y)$~~ ^{$l(a, y)$} for action $a \in A$, outcome $y \in \{0, 1\}$

A best response function ~~$BR_x(p)$~~ ^{$BR_\ell(p)$} = ~~$\arg\max$~~ ^{$\arg\max$} ~~$U(a, y)$~~ ^{$l(a, y)$} $a \in A \quad y \sim \text{Ber}(p)$

Definition: For distribution D , actions A , and collection of losses $\mathcal{L}$, and collection of predictors $\mathcal{H} = \{h: \mathcal{X} \rightarrow [0, 1]\}$, f is an α -omnipredictor if for every $\ell \in \mathcal{L}$, every $h \in \mathcal{H}$

$$\mathbb{E}_D[\ell(BR_\ell(f(x)), y)] \leq \mathbb{E}_D[\ell(BR_\ell(h(x)), y)] + \alpha$$

Minimizing MSE $\Rightarrow$ Omniprediction

$$\begin{array}{l} \text{domain} \\ \mathcal{X} = \{x_1, x_2\} \end{array} \quad \begin{array}{l} \text{model class} \\ \mathcal{H} = \{h_t : t \in [0, 1]\} \end{array} \quad h_t(x) = \begin{cases} t & \text{if } x = x_1 \\ t/2 & \text{if } x = x_2 \end{cases}$$

$$\begin{array}{l} \text{distribution} \\ \mathbb{P}[Y=1 | X=x_1] = .4 \quad \mathbb{P}[Y=1 | X=x_2] = .6 \quad \mathbb{P}[X=x_1] = \mathbb{P}[X=x_2] = .5 \end{array}$$

Minimizing $\text{MSE}(h_t) \propto (t-.4)^2 + (t/2-.6)^2$ gives $t^* = .56$

Consider absolute loss $\ell(y, a) = |y - a|$ $\text{BR}_\ell(p) = \begin{cases} 1 & \text{if } p \geq .5 \\ 0 & \text{if } p < .5 \end{cases}$

Utility is higher if you do $\text{BR}_\ell(h_{.4999}(x))$ than $\text{BR}_\ell(h_{t^*}(x))$